

Absolutely. Since this is **your own project**, there's no point in studying every deep learning topic under the sun. Instead, study **only the concepts that are actually used in your DocVision-AI project**. This will make you interview-ready because you'll be able to explain every line of your code.

I recommend studying the topics in this exact order:

1. Artificial Neural Network (ANN)

Definition

An Artificial Neural Network (ANN) is a computational model inspired by the human brain. It consists of neurons arranged in layers that learn patterns from data.

Why do we need it?

ANNs can learn relationships between inputs and outputs. CNNs are actually a specialized type of ANN.

Example

Suppose you want to predict whether a student will pass based on:

- Study Hours
- Attendance

Input:

[8 Hours, 90%]



Output:

Pass



In Our Project

The Dense layers at the end of the CNN are ANN layers that classify the extracted image features into **Resume** or **Invoice**.

2. Convolutional Neural Network (CNN)

Definition

A CNN is a deep learning model designed specifically for images. It automatically learns visual features such as edges, shapes, logos, tables, and text layouts.

Why do we need it?

A normal ANN would require millions of parameters for images, while CNNs reuse filters and preserve spatial information.

Example

For a cat image:

- Layer 1 learns edges.
- Layer 2 learns ears.
- Layer 3 learns face.
- Final layer predicts "Cat."

In Our Project

The CNN learns the layout of resumes and invoices instead of manually engineered features.

3. Input Image

Definition

An image is converted into a tensor before being fed into the model.

Example:

`224 × 224 × 3`



means

`Height = 224`
`Width = 224`
`Channels = 3 (RGB)`



In Our Project

`Resume Image`



`↓`

`224×224×3 Tensor`

↓

CNN

4. Image Preprocessing

Definition

Preparing images before training.

Includes:

- Resize
- Padding
- RGB conversion
- Normalization

Example

Original

1000×700



↓

Resize + White Padding

↓

224×224

Why?

Every image must have the same size.

5. Normalization

Definition

Converting pixel values from

0-255



to

0-1



Formula

`pixel / 255`



Example

128



↓

128/255

↓

0.50

Why?

Helps the optimizer converge faster.

6. Data Augmentation

Definition

Artificially creating new training images.

Example

Original Resume

↓

Rotate

↓

Zoom

↓

Contrast

↓

Translate

Why?

Reduces overfitting.

In Our Project

We used

- Rotation
- Zoom
- Translation
- Contrast

No horizontal flip because documents shouldn't be mirrored.

7. Convolution Layer

Definition

A convolution layer slides a small filter across the image to extract features.

Example

3×3 Kernel

```
1 0 1
0 1 0
1 0 1
```



Moves across the image.

Learns

- Edges
 - Shapes
 - Text regions
 - Tables
-

8. Kernel / Filter

Definition

A small matrix that detects patterns.

Example

3×3



5×5

In Our Project

The CNN learns filters automatically.

9. Feature Map

Definition

The output produced after convolution.

Instead of the original image,

the CNN now sees

Edges



Corners

Patterns

10. ReLU

Definition

Activation Function

Formula

$\text{ReLU}(x) = \max(0, x)$



Example

Input

-4

2



8
-1

Output

0
2
8
0



Why?

Introduces non-linearity.

11. MaxPooling

Definition

Reduces image size while keeping important information.

Example

4 2
8 6



↓

8



Why?

Reduces computation.

Reduces overfitting.

12. Layer Normalization

Definition

Normalizes the activations within each sample independently to stabilize learning.

Why?

Makes training more stable.

In Our Project

We replaced BatchNormalization during debugging experiments.

13. Global Average Pooling

Definition

Instead of flattening the feature maps, it averages each feature map into a single value.

Example

Feature Map

2	4
6	8



Average

5



Why?

Reduces parameters.

Prevents overfitting.

14. Dense Layer

Definition

Fully connected layer.

Makes the final decision.

In Our Project

Learns whether the extracted features belong to

Resume

or

Invoice.

15. Dropout

Definition

Randomly disables neurons during training.

Example

100 neurons



↓

Dropout (0.5)

↓

Only 50 remain active

Why?

Prevents overfitting.

16. Softmax

Definition

Converts raw outputs into probabilities.

Example

Resume



2.4

Invoice

1.1

↓

Softmax

Resume



78%

Invoice

22%

All probabilities sum to

1



17. Forward Propagation

Definition

The image moves through the network.

Image



↓

CNN

↓

Pooling

↓

Dense

↓

Prediction

18. Loss Function

Definition

Measures how wrong the prediction is.

In Our Project



Because

Resume



0

Invoice

1

are integer labels.

19. Backpropagation

Definition

Updates the weights using the error produced by the loss function.

Simple flow:

Prediction



↓

Loss

↓

Gradient

↓

Update Weights

20. Gradient Descent

Definition

Optimization algorithm that reduces the loss.

Think of it like walking downhill until you reach the lowest point.

21. Optimizer (Adam)

Definition

Algorithm that updates weights efficiently.

Why Adam?

- Fast convergence
 - Adaptive learning rate
 - Works well for most deep learning problems
-

22. Learning Rate

Definition

Controls the size of each weight update.

Too high → overshoots the optimum.

Too low → training becomes very slow.

23. Epoch

Definition

One complete pass through the entire training dataset.

Example

1252 images

↓

1 Epoch

means all 1252 images have been seen once.

24. Batch Size

Definition

Number of samples processed before updating the weights.

Example

Batch Size

32



1252 images



40 mini-batches approximately.

25. Training vs Validation vs Test

Training

Model learns.

Validation

Model is tuned.

Test

Final unbiased evaluation.

26. Overfitting

Definition

The model memorizes the training data but performs poorly on unseen data.

Example

Train Accuracy



99%

Validation

60%

27. EarlyStopping

Definition

Stops training when validation performance stops improving.

Why?

Prevents overfitting and saves time.

28. Model Checkpoint

Definition

Automatically saves the best model during training.

In your project:

```
best_model.keras
```



stores the weights from the epoch with the best validation performance.

29. Model Evaluation

You used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

These tell you **how well** the model performs, not just whether it is right or wrong overall.

30. Inference

Definition

Using the trained model to make predictions on new, unseen data.

Flow:

Image



↓

Preprocessing

↓

Load Model

↓

Predict

↓

Softmax

↓

Resume / Invoice

★ Final Advice

These **30 topics** are the **core deep learning concepts directly used in your DocVision-AI project**. If you can confidently explain each one—definition, why it's needed, a simple example, and how it fits into your project—you'll be well prepared to discuss your work in interviews.

I recommend studying them in this order:

1. ANN
2. CNN
3. Image Preprocessing
4. Convolution → ReLU → Pooling
5. Dense → Softmax
6. Loss → Backpropagation → Optimizer
7. Epochs → Batch Size → Learning Rate
8. Overfitting → Dropout → EarlyStopping
9. Evaluation Metrics
10. Inference Pipeline

Mastering these topics will let you explain not only **how** your model works, but also **why** you designed it the way you did.